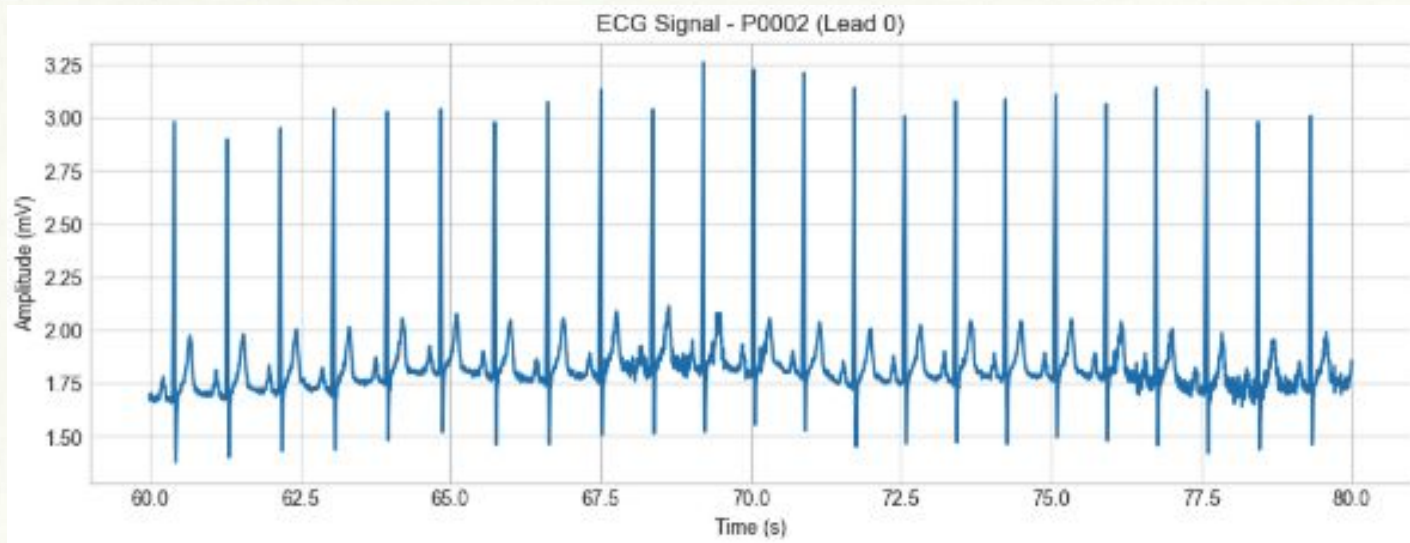
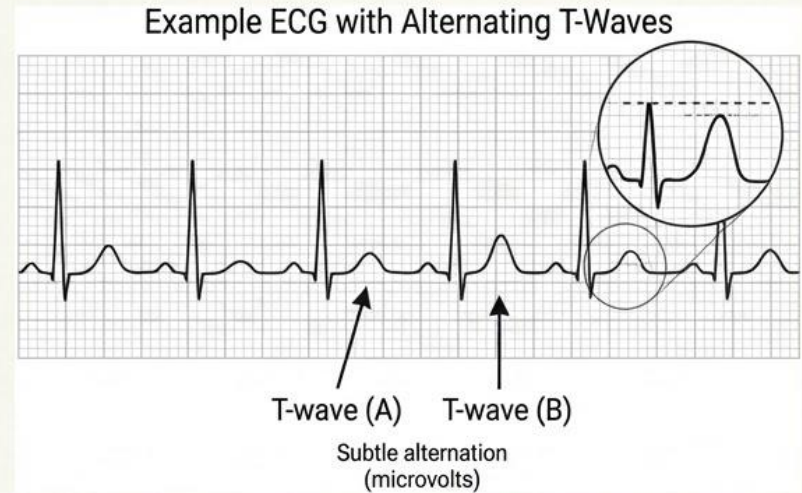


Correlation Method for Detection of Transient T-Wave Alternans in Digital Holter ECG Recordings



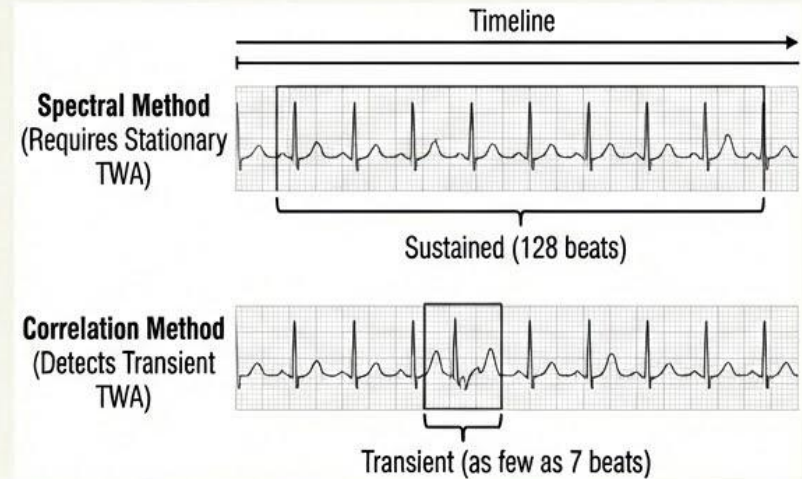
What is T-Wave Alternans?

- **Definition:** Beat-to-beat alternation in T-wave morphology
- **Clinical Significance:** Marker of myocardial electrical instability
- **Challenge:** Microvolt-level changes (invisible to naked eye)
- **Goal:** Detect TWA in ambulatory Holter ECG recordings

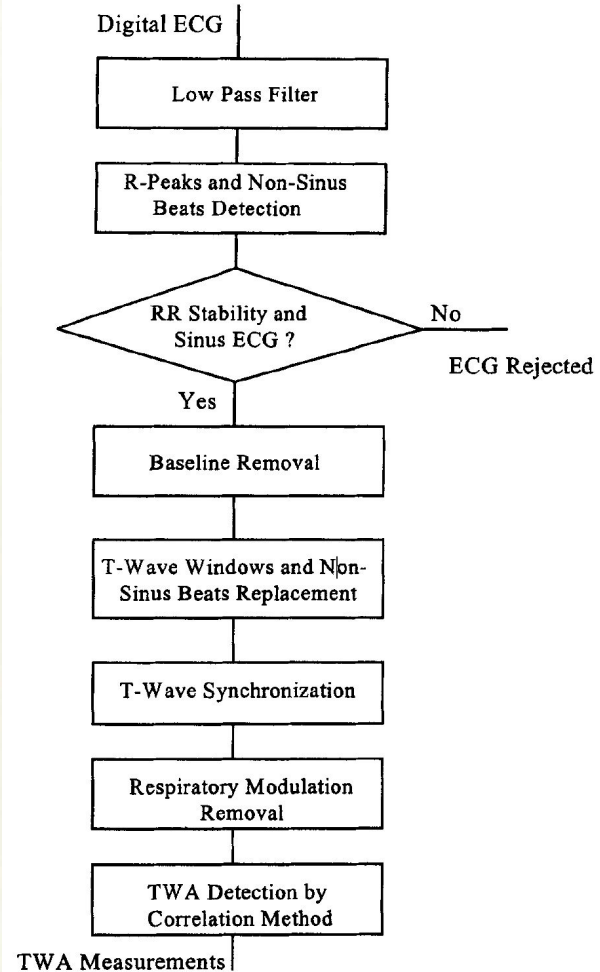


The Correlation Method (Burattini 1999)

- **Key Innovation:** Time-domain approach that detects:
 - Stationary TWA (sustained, 128 beats)
 - Nonstationary TWA (transient, as few as 7 beats)
- **Advantage over Spectral Method:**
 - Better detection of transient episodes
 - Works with sinus rhythm ECGs



Algorithm Pipeline



Key Equations

Alternans Correlation Index (ACI): $ACI_j = \frac{\sum T_j(i) \cdot T_{mdn}(i)}{\sum T_{mdn}^2(i)}$

Alternans Amplitude (ACM): $A_{CM}(j) = 2|ACI_j - 1| \cdot \frac{\sum T_{mdn}^2(i)}{\sum |T_{mdn}(i)|}$

TWA Magnitude: $MAG_{CM} = A_{CM} \times N_{CM}$

Threshold: $MAG > 550 \mu V \rightarrow$ TWA Positive

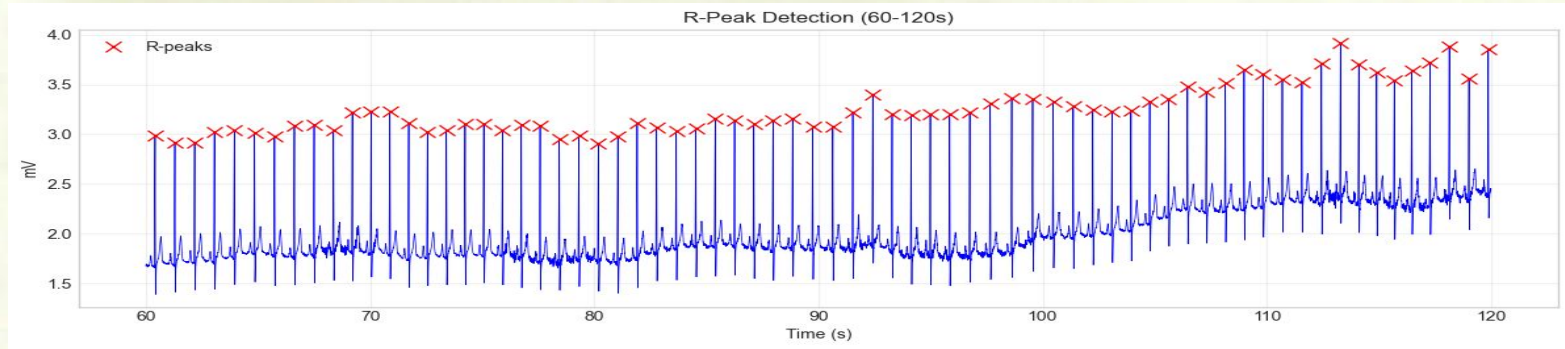
Dataset

Group	Database	Patients
SCD	MUSIC (Sudden Cardiac Death)	8

- Sampling: 128-250 Hz
- Duration: 24-hour Holter recordings
- Analysis: Best 128-beat window per patient

ECG Preprocessing

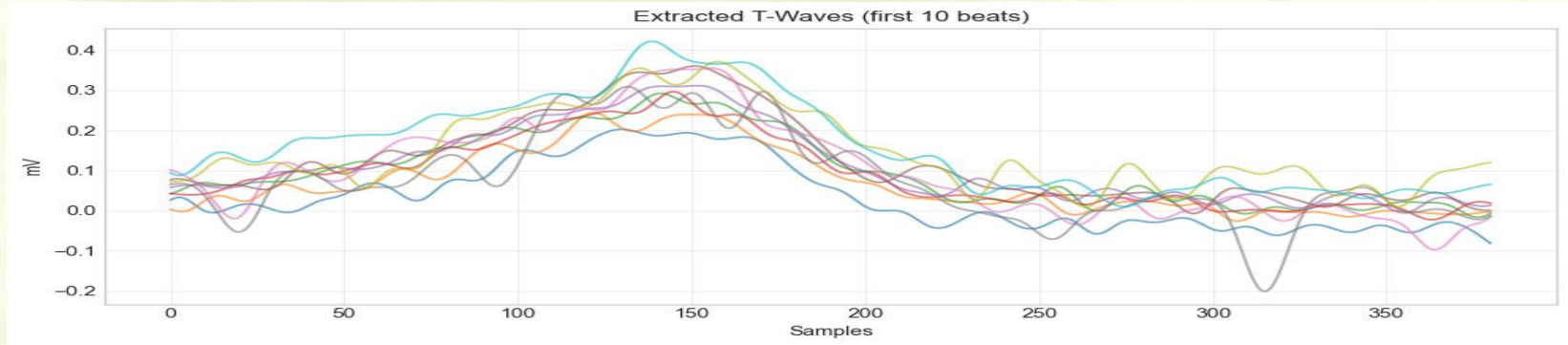
Patient P0002 - Step 1: Signal Filtering & R-Peak Detection



- Low-pass filtered at 60 Hz
- R-peaks detected using maximum derivative method
- Detected 94,890 R-peaks (73.3 BPM)

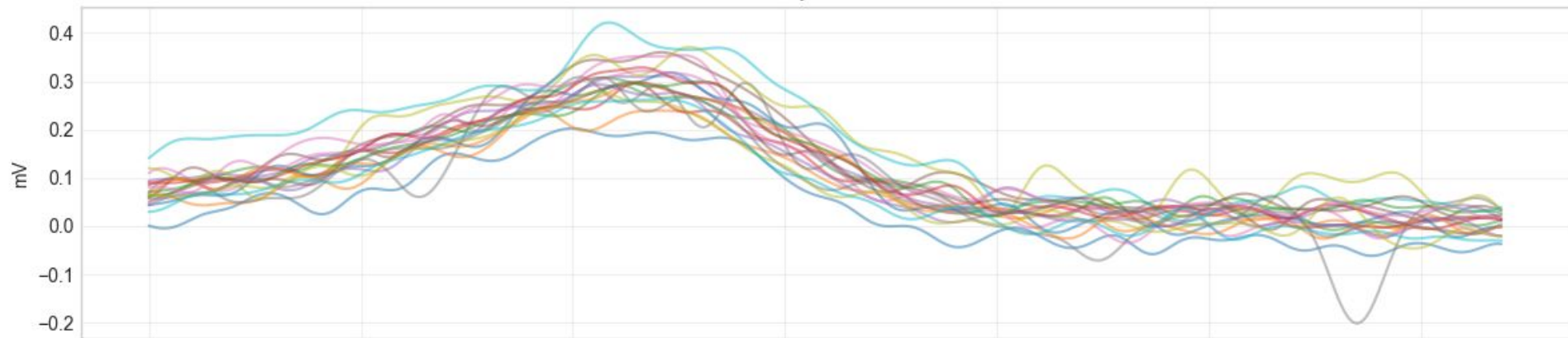
T-Wave Extraction & Alignment

Patient P0002 - Step 2: T-Wave Processing

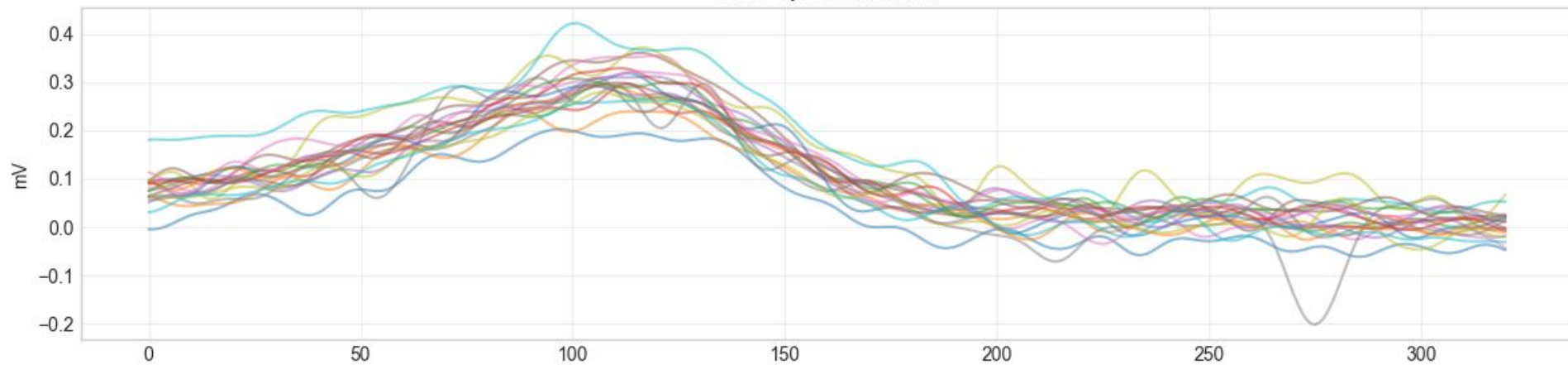


- T-wave window: Heart rate-dependent onset + duration
- Synchronized using $\pm 30\text{ms}$ cross-correlation
- Respiratory modulation removed (0.14-0.35 c/b band-stop)

Before Synchronization



After Synchronization



TWA Detection Result

Patient P0002

File Validation (All Windows)

Metric	Value
Median RR Variability	4.5%
Windows Accepted	879/945 (93%)

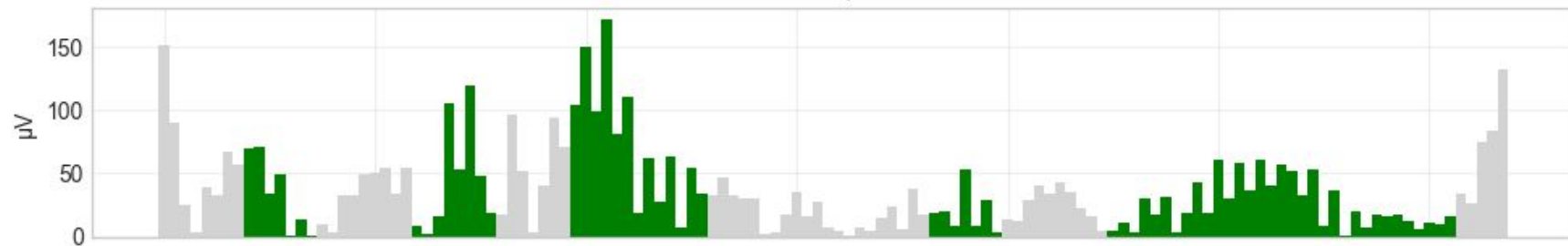
Single Window Analysis [0]

Metric	Value
Alternating Beats	68
Mean Amplitude	37.8 μ V
TWA Magnitude	2571 μV → POSITIVE

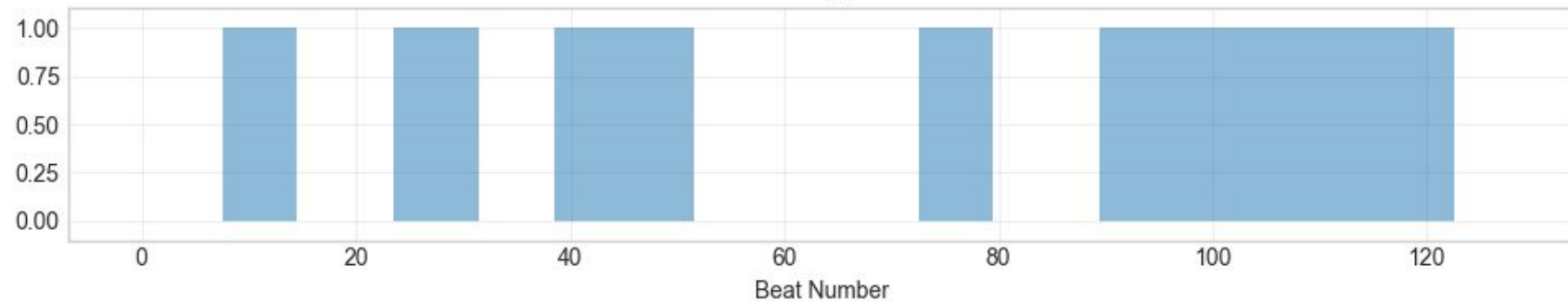
Alternans Correlation Index



TWA Amplitude

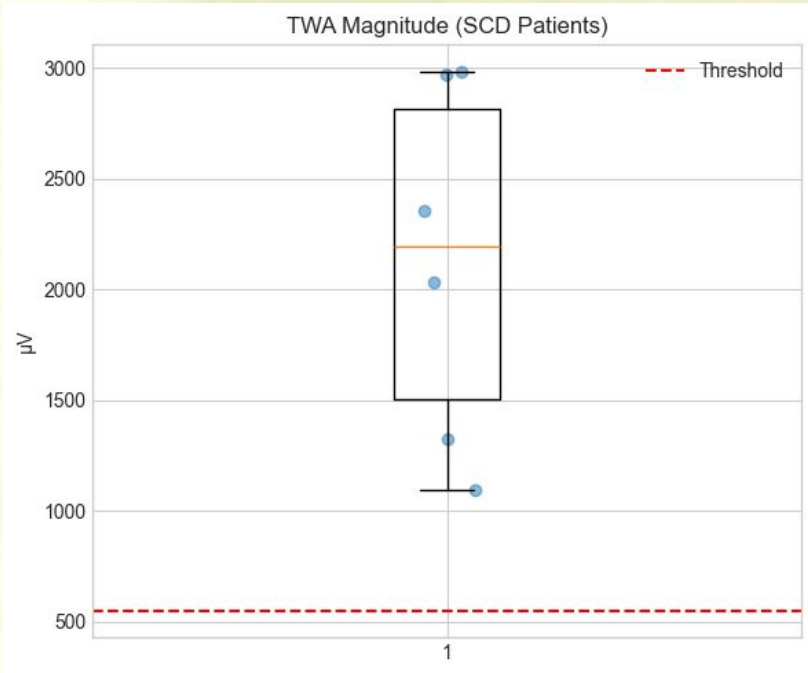


Alternating Beats



SCD Group Results

Analyzing 8 SCD Patients



Metric	SCD Group
Patients Analyzed	6
Mean Magnitude	2126 µV
Max Magnitude	2982 µV
Positive Rate	6/6 (100%)

Limitations

- RR stability requirement ($\leq 10\%$) excludes arrhythmic patients
- Small sample size (8 SCD)
- Single-lead analysis only
- High detection rate may indicate sensitivity to noise

Conclusions

- Successfully implemented Burattini Correlation Method
- Tested on MIT-BIH Normal and MUSIC SCD databases

